

Q1. Two vertices of a triangle ABC are $A(3, -1)$ and $B(-2, 3)$, and its orthocentre is $P(1, 1)$. If the coordinates of the point C are (α, β) and the centre of the circle circumscribing the triangle PAB is (h, k) , then the value of $(\alpha + \beta) + 2(h + k)$ equals _____ **09 Apr 2024 (E)**

- (1) 5 (2) 81
(3) 15 (4) 51

Q2. A ray of light coming from the point $P(1, 2)$ gets reflected from the point Q on the x -axis and then passes through the point $R(4, 3)$. If the point $S(h, k)$ is such that PQRS is a parallelogram, then hk^2 is equal to : **09 Apr 2024 (M)**

- (1) 70 (2) 80
(3) 60 (4) 90

Q3. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is _____ **08 Apr 2024 (M)**

Q4. If the image of the point $(-4, 5)$ in the line $x + 2y = 2$ lies on the circle $(x + 4)^2 + (y - 3)^2 = r^2$, then r is equal to: **08 Apr 2024 (E)**

- (1) 2 (2) 3
(3) 1 (4) 4

Q5. The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $(2, -\frac{4}{3})$ divides the third side BC internally in the ratio 2 : 1. the equation of the side BC is **08 Apr 2024 (M)**

- (1) $x + 3y + 2 = 0$ (2) $x - 6y - 10 = 0$
(3) $x - 3y - 6 = 0$ (4) $x + 6y + 6 = 0$

Q6. If the line segment joining the points $(5, 2)$ and $(2, a)$ subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is : **08 Apr 2024 (E)**

- (1) 6 (2) 8
(3) 2 (4) -4

Q7. Let a ray of light passing through the point $(3, 10)$ reflects on the line $2x + y = 6$ and the reflected ray passes through the point $(7, 2)$. If the equation of the incident ray is $ax + by + 1 = 0$, then $a^2 + b^2 + 3ab$ is equal to _____ **08 Apr 2024 (E)**

Q8. If $A(3, 1, -1)$, $B(\frac{5}{3}, \frac{7}{3}, \frac{1}{3})$, $C(2, 2, 1)$ and $D(\frac{10}{3}, \frac{2}{3}, \frac{-1}{3})$ are the vertices of a quadrilateral ABCD, then its area is **06 Apr 2024 (M)**

- (1) $\frac{2\sqrt{2}}{3}$ (2) $\frac{5\sqrt{2}}{3}$
(3) $2\sqrt{2}$ (4) $\frac{4\sqrt{2}}{3}$

Q9. If $P(6, 1)$ be the orthocentre of the triangle whose vertices are $A(5, -2)$, $B(8, 3)$ and $C(h, k)$, then the point C lies on the circle: **06 Apr 2024 (E)**

- (1) $x^2 + y^2 - 61 = 0$ (2) $x^2 + y^2 - 52 = 0$
(3) $x^2 + y^2 - 65 = 0$ (4) $x^2 + y^2 - 74 = 0$

Q10. Let a variable line of slope $m > 0$ passing through the point $(4, -9)$ intersect the coordinate axes at the points A and B . The minimum value of the sum of the distances of A and B from the origin is 06 Apr 2024 (M)

- (1) 30 (2) 25
(3) 15 (4) 10

Q11. If the locus of the point, whose distances from the point $(2, 1)$ and $(1, 3)$ are in the ratio $5 : 4$, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to : 06 Apr 2024 (E)

- (1) 37 (2) 437
(3) -27 (4) 5

Q12. If $A(1, -1, 2)$, $B(5, 7, -6)$, $C(3, 4, -10)$ and $D(-1, -4, -2)$ are the vertices of a quadrilateral $ABCD$, then its area is : 05 Apr 2024 (M)

- (1) $48\sqrt{7}$ (2) $12\sqrt{29}$
(3) $24\sqrt{7}$ (4) $24\sqrt{29}$

Q13. Let two straight lines drawn from the origin O intersect the line $3x + 4y = 12$ at the points P and Q such that $\triangle OPQ$ is an isosceles triangle and $\angle POQ = 90^\circ$. If $l = OP^2 + PQ^2 + QO^2$, then the greatest integer less than or equal to l is : 05 Apr 2024 (M)

- (1) 42 (2) 46
(3) 44 (4) 48

Q14. Let $A(-1, 1)$ and $B(2, 3)$ be two points and P be a variable point above the line AB such that the area of $\triangle PAB$ is 10. If the locus of P is $ax + by = 15$, then $5a + 2b$ is : 05 Apr 2024 (E)

- (1) 6 (2) $-\frac{6}{5}$
(3) 4 (4) $-\frac{12}{5}$

Q15. The vertices of a triangle are $A(-1, 3)$, $B(-2, 2)$ and $C(3, -1)$. A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is : 04 Apr 2024 (M)

- (1) $x + y + (2 - \sqrt{2}) = 0$ (2) $-x + y - (2 - \sqrt{2}) = 0$
(3) $x + y - (2 - \sqrt{2}) = 0$ (4) $x - y - (2 + \sqrt{2}) = 0$

Q16. Consider a triangle ABC having the vertices $A(1, 2)$, $B(\alpha, \beta)$ and $C(\gamma, \delta)$ and angles $\angle ABC = \frac{\pi}{6}$ and $\angle BAC = \frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to 04 Apr 2024 (E)

Q17. Let ABC be an isosceles triangle in which A is at $(-1, 0)$, $\angle A = \frac{2\pi}{3}$, $AB = AC$ and B is on the positive x -axis. If $BC = 4\sqrt{3}$ and the line BC intersects the line $y = x + 3$ at (α, β) , then $\frac{\beta^4}{\alpha^2}$ is: 01 Feb 2024 (E)

Q18. Let $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and let $A(\alpha, \beta)$, $B(1, 0)$, $C(\gamma, \delta)$ and $D(1, 2)$ be the vertices of a parallelogram $ABCD$. If $AB = \sqrt{10}$ and the points A and C lie on the line $3y = 2x + 1$, then $2(\alpha + \beta + \gamma + \delta)$ is equal to 31 Jan 2024 (M)

- (1) 10 (2) 5
(3) 12 (4) 8

Q19. Let $A(a, b)$, $B(3, 4)$ and $(-6, -8)$ respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point $P(2a + 3, 7b + 5)$ from the line $2x + 3y - 4 = 0$ measured parallel to the line $x - 2y - 1 = 0$ is

31 Jan 2024 (E)

(1) $\frac{15\sqrt{5}}{7}$

(2) $\frac{17\sqrt{5}}{6}$

(3) $\frac{17\sqrt{5}}{7}$

(4) $\frac{\sqrt{5}}{17}$

Q20. Let $A(-2, -1)$, $B(1, 0)$, $C(\alpha, \beta)$ and $D(\gamma, \delta)$ be the vertices of a parallelogram $ABCD$. If the point C lies on $2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

31 Jan 2024 (E)

Q21. A line passing through the point $A(9, 0)$ makes an angle of 30° with the positive direction of x -axis. If this line is rotated about A through an angle of 15° in the clockwise direction, then its equation in the new position is

30 Jan 2024 (M)

(1) $\frac{y}{\sqrt{3}-2} + x = 9$

(2) $\frac{x}{\sqrt{3}-2} + y = 9$

(3) $\frac{x}{\sqrt{3}+2} + y = 9$

(4) $\frac{y}{\sqrt{3}+2} + x = 9$

Q22. If $x^2 - y^2 + 2hxy + 2gx + 2fy + c = 0$ is the locus of a point, which moves such that it is always equidistant from the lines $x + 2y + 7 = 0$ and $2x - y + 8 = 0$, then the value of $g + c + h - f$ equals

30 Jan 2024 (E)

(1) 14

(2) 6

(3) 8

(4) 29

Q23. Let A be the point of intersection of the lines $3x + 2y = 14$, $5x - y = 6$ and B be the point of intersection of the lines $4x + 3y = 8$, $6x + y = 5$. The distance of the point $P(5, -2)$ from the line AB is

29 Jan 2024 (E)

(1) $\frac{13}{2}$

(2) 8

(3) $\frac{5}{2}$

(4) 6

Q24. Let $(5, \frac{a}{4})$, be the circumcenter of a triangle with vertices $A(a, -2)$, $B(a, 6)$ and $C(\frac{a}{4}, -2)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is

29 Jan 2024 (M)

(1) 60

(2) 53

(3) 62

(4) 30

Q25. The distance of the point $(2, 3)$ from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to

29 Jan 2024 (E)

(1) $4\sqrt{2}$

(2) $6\sqrt{3}$

(3) $3 + 4\sqrt{2}$

(4) $4 + 6\sqrt{3}$

Q26*. In a ΔABC , suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the point A and B are respectively $(4, 6)$ and (α, β) , then $\alpha + 2\beta$ is equal to

29 Jan 2024 (M)

(1) -4

(2) 42

(3) 2

(4) -1

Q27. Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B . Then the distance of the point $P(m, n)$ from the point $Q(-2, -3)$ is

27 Jan 2024 (E)

(1) 10

(3) 4

(2) 6

(4) 8

Q28. If the sum of squares of all real values of α , for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p , then the greatest integer less than or equal to p is _____.

27 Jan 2024 (E)

Q29. Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a , for which the points $(a^2, a + 1)$ lie in R , is :

27 Jan 2024 (E)

(1) $(-3, -1) \cup (-\frac{1}{3}, 1)$ (2) $(-3, 0) \cup (\frac{1}{3}, 1)$ (3) $(-3, 0) \cup (\frac{2}{3}, 1)$ (4) $(-3, -1) \cup (\frac{1}{3}, 1)$

Q30. The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is :

27 Jan 2024 (M)

(1) $\frac{8}{5}$ (2) $\frac{25}{41}$ (3) $\frac{2}{5}$ (4) $\frac{30}{41}$

Q31. If (α, β) is the orthocenter of the triangle ABC with vertices $A(3, -7)$, $B(-1, 2)$ and $C(4, 5)$, then $9\alpha - 6\beta + 60$ is equal to

15 Apr 2023 (M)

(1) 25

(2) 35

(3) 30

(4) 40

Q32. Consider the triangles with vertices $A(2, 1)$, $B(0, 0)$ and $C(t, 4)$, $t \in [0, 4]$. If the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively, then $6\alpha + 21\beta$ is equal to _____.

15 Apr 2023 (M)

Q33. Let (α, β) be the centroid of the triangle formed by the lines $15x - y = 82$, $6x - 5y = -4$ and $9x + 4y = 17$. Then $\alpha + 2\beta$ and $2\alpha - \beta$ are the roots of the equation

13 Apr 2023 (E)

(1) $x^2 - 7x + 12 = 0$ (2) $x^2 - 14x + 48 = 0$ (3) $x^2 - 13x + 42 = 0$ (4) $x^2 - 10x + 25 = 0$

Q34. If the point $(\alpha, \frac{7\sqrt{3}}{3})$ lies on the curve traced by the mid-points of the line segments of the lines $x \cos \theta + y \sin \theta = 7$, $\theta \in (0, \frac{\pi}{2})$ between the co-ordinates axes, then α is equal to

12 Apr 2023 (M)

(1) -7 (2) $-7\sqrt{3}$ (3) $7\sqrt{3}$ (4) 7

Q35*. If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2 - \alpha - \beta$ is equal to

11 Apr 2023 (E)

Q36. Let the equations of two adjacent sides of a parallelogram $ABCD$ be $2x - 3y = -23$ and $5x + 4y = 23$. If the equation of its one diagonal AC is $3x + 7y = 23$ and the distance of A from the other diagonal is d , then $50d^2$ is equal to _____

10 Apr 2023 (E)

Q37. Let $C(\alpha, \beta)$ be the circumcentre of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$. Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to 08 Apr 2023 (M)

- (1) 18 (2) 17
(3) 15 (4) 16

Q38. The straight lines l_1 and l_2 pass through the origin and trisect the line segment of the line $L : 9x + 5y = 45$ between the axes. If m_1 and m_2 are the slopes of the lines l_1 and l_2 , then the point of intersection of the line $y = (m_1 + m_2)x$ with L lies on 06 Apr 2023 (M)

- (1) $y - 2x = 5$ (2) $6x + y = 10$
(3) $y - x = 5$ (4) $6x - y = 15$

Q39. If the orthocentre of the triangle, whose vertices are $(1, 2)$, $(2, 3)$ and $(3, 1)$ is (α, β) , then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is 01 Feb 2023 (M)

- (1) $x^2 - 19x + 90 = 0$ (2) $x^2 - 18x + 80 = 0$
(3) $x^2 - 22x + 120 = 0$ (4) $x^2 - 20x + 99 = 0$

Q40*. The combined equation of the two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ can be written as $(ax + by + c)(a'x + b'y + c') = 0$. The equation of the angle bisectors of the lines represented by the equation $2x^2 + xy - 3y^2 = 0$ is 01 Feb 2023 (M)

- (1) $3x^2 + 5xy + 2y^2 = 0$ (2) $x^2 - y^2 + 10xy = 0$
(3) $3x^2 + xy - 2y^2 = 0$ (4) $x^2 - y^2 - 10xy = 0$

Q41. A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive directions of x -axis and y -axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y -axis and the area of $\triangle OAB$ is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to: 30 Jan 2023 (M)

- (1) $\frac{392}{3}$ (2) 196
(3) $\frac{196}{3}$ (4) 98

Q42. Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that $\triangle ABC$ is an equilateral triangle. Then, the area of the $\triangle ABC$ is 29 Jan 2023 (M)

- (1) $3\sqrt{3}$ (2) $2\sqrt{3}$
(3) $\frac{8}{\sqrt{3}}$ (4) $\frac{10}{\sqrt{3}}$

Q43. A light ray emits from the origin making an angle 30° with the positive x -axis. After getting reflected by the line $x + y = 1$, if this ray intersects x -axis at Q , then the abscissa of Q is 29 Jan 2023 (M)

- (1) $\frac{2}{(\sqrt{3}-1)}$ (2) $\frac{2}{3+\sqrt{3}}$
(3) $\frac{2}{3-\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2(\sqrt{3}+1)}$

Q44. A triangle is formed by X -axis, Y -axis and the line $3x + 4y = 60$. Then the number of points $P(a, b)$ which lie strictly inside the triangle, where a is an integer and b is a multiple of a , is 25 Jan 2023 (E)

Q45. The equations of the sides AB , BC & CA of a triangle ABC are $2x + y = 0$, $x + py = 21a$ ($a \neq 0$) and $x - y = 3$ respectively. Let $P(2, a)$ be the centroid of the triangle ABC , then $(BC)^2$ is equal to

24 Jan 2023 (E)

Q46. Let $A(\alpha, -2)$, $B(\alpha, 6)$ and $C(\frac{\alpha}{4}, -2)$ be vertices of a $\triangle ABC$. If $(5, \frac{\alpha}{4})$ is the circumcentre of $\triangle ABC$, then which of the following is NOT correct about $\triangle ABC$

29 Jul 2022 (E)

- (1) area is 24 (2) perimeter is 25
(3) circumradius is 5 (4) inradius is 2

Q47. Let the circumcentre of a triangle with vertices $A(a, 3)$, $B(b, 5)$ and $C(a, b)$, $ab > 0$ be $P(1, 1)$. If the line AP intersects the line BC at the point $Q(k_1, k_2)$, then $k_1 + k_2$ is equal to

29 Jul 2022 (M)

- (1) 2 (2) $\frac{4}{7}$
(3) $\frac{2}{7}$ (4) 4

Q48. Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3(m_1^2 + m_2^2) = 220$. If one vertex of the square is $(10(\cos \alpha - \sin \alpha), 10(\sin \alpha + \cos \alpha))$, where $\alpha \in (0, \frac{\pi}{2})$ and the equation of one diagonal is $(\cos \alpha - \sin \alpha)x + (\sin \alpha + \cos \alpha)y = 10$, then $72(\sin^4 \alpha + \cos^4 \alpha) + a^2 - 3a + 13$ is equal to

29 Jul 2022 (E)

- (1) 119 (2) 128
(3) 145 (4) 155

Q49. For $t \in (0, 2\pi)$, if ABC is an equilateral triangle with vertices $A(\sin t, -\cos t)$, $B(\cos t, \sin t)$ and $C(a, b)$ such that its orthocentre lies on a circle with centre $(1, \frac{1}{3})$, then $(a^2 - b^2)$ is equal to

28 Jul 2022 (M)

- (1) $\frac{8}{3}$ (2) 8
(3) $\frac{77}{9}$ (4) $\frac{80}{9}$

Q50. Let $A(1, 1)$, $B(-4, 3)$, $C(-2, -5)$ be vertices of a triangle ABC , P be a point on side BC , and Δ_1 and Δ_2 be the areas of triangle APB and ABC . Respectively. If $\Delta_1 : \Delta_2 = 4 : 7$, then the area enclosed by the lines AP , AC and the x -axis is

27 Jul 2022 (M)

- (1) $\frac{1}{4}$ (2) $\frac{3}{4}$
(3) $\frac{1}{2}$ (4) 1

Q51. The equations of the sides AB , BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 39$ and $x - y = 3$ respectively and $P(2, 3)$ is its circumcentre. Then which of the following is NOT true

27 Jul 2022 (E)

- (1) $(AC)^2 = 9p$ (2) $(AC)^2 + p^2 = 136$
(3) $32 < \text{area}(\triangle ABC) < 36$ (4) $34 < \text{area}(\triangle ABC) < 38$

Q52. A point P moves so that the sum of squares of its distances from the points $(1, 2)$ and $(-2, 1)$ is 14. Let $f(x, y) = 0$ be the locus of P , which intersects the x -axis at the points A, B and the y -axis at the point C, D .

Then the area of the quadrilateral $ACBD$ is equal to

26 Jul 2022 (M)

- (1) $\frac{9}{2}$ (2) $\frac{3\sqrt{17}}{2}$
(3) $\frac{3\sqrt{17}}{4}$ (4) 9

Q53. The equations of the sides AB , BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is $(2, a)$, $-\frac{1}{2} < a < 2$, then p is equal to

26 Jul 2022 (M)

Q54. A line, with the slope greater than one, passes through the point $A(4, 3)$ and intersects the line $x - y - 2 = 0$ at the point B . If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line **25 Jul 2022 (M)**

(1) $2x + y = 9$

(2) $3x - 2y = 7$

(3) $x + 2y = 6$

(4) $2x - 3y = 3$

Q55. Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2 , then $100(\alpha + \beta)$ is equal to **25 Jul 2022 (E)**

(1) -14

(2) 42

(3) -22

(4) 14

Q56. The distance of the origin from the centroid of the triangle whose two sides have the equations $x - 2y + 1 = 0$ and $2x - y - 1 = 0$ and whose orthocenter is $(\frac{7}{3}, \frac{7}{3})$ is: **29 Jun 2022 (E)**

(1) $\sqrt{2}$

(2) 2

(3) $2\sqrt{2}$

(4) 4

Q57. The distance between the two points A and A' which lie on $y = 2$ such that both the line segments AB and $A'B$ (where B is the point $(2, 3)$) subtend angle $\frac{\pi}{4}$ at the origin, is equal to **29 Jun 2022 (M)**

(1) 10

(2) $\frac{48}{5}$

(3) $\frac{52}{5}$

(4) 3

Q58. Let a triangle be bounded by the lines $L_1 : 2x + 5y = 10$; $L_2 : -4x + 3y = 12$ and the line L_3 , which passes through the point $P(2, 3)$, intersect L_2 at A and L_1 at B . If the point P divides the line-segment AB , internally in the ratio $1 : 3$, then the area of the triangle is equal to **28 Jun 2022 (E)**

(1) $\frac{110}{13}$

(2) $\frac{132}{13}$

(3) $\frac{142}{13}$

(4) $\frac{151}{13}$

Q59. A ray of light passing through the point $P(2, 3)$ reflects on the X -axis at point A and the reflected ray passes through the point $Q(5, 4)$. Let R be the point that divides the line segment AQ internally into the ratio $2 : 1$. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be (α, β) . Then, the value of $7\alpha + 3\beta$ is equal to _____. **28 Jun 2022 (M)**

Q60. In an isosceles triangle ABC , the vertex A is $(6, 1)$ and the equation of the base BC is $2x + y = 4$. Let the point B lie on the line $x + 3y = 7$. If (α, β) is the centroid ΔABC , then $15(\alpha + \beta)$ is equal to **27 Jun 2022 (M)**

(1) 51

(2) 39

(3) 41

(4) 49

Q61. Let R be the point $(3, 7)$ and let P and Q be two points on the line $x + y = 5$ such that PQR is an equilateral triangle. Then the area of ΔPQR is **26 Jun 2022 (M)**

(1) $\frac{25}{4\sqrt{3}}$

(2) $\frac{25\sqrt{3}}{2}$

(3) $\frac{25}{\sqrt{3}}$

(4) $\frac{25}{2\sqrt{3}}$

Q62. Let the area of the triangle with vertices $A(1, \alpha)$, $B(\alpha, 0)$ and $C(0, \alpha)$ be 4 sq. units. If the points $(\alpha, -\alpha)$, $(-\alpha, \alpha)$ and (α^2, β) are collinear, then β is equal to **24 Jun 2022 (E)**

- (1) 64 (2) -8
(3) -64 (4) 512

Q63. Let $A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right)$, $a > 0$, be a fixed point in the xy -plane. The image of A in y -axis be B and the image of B in x -axis be C . If $D(3 \cos \theta, a \sin \theta)$, is a point in the fourth quadrant such that the maximum area of $\triangle ACD$ is 12 square units, then a is equal to _____ **24 Jun 2022 (M)**

Q64. Let the points of intersections of the lines $x - y + 1 = 0$, $x - 2y + 3 = 0$ and $2x - 5y + 11 = 0$ are the mid points of the sides of a triangle ABC . Then the area of the triangle ABC is _____ **01 Sep 2021 (E)**

Q65. A man starts walking from the point $P(-3, 4)$, touches the x -axis at R , and then turns to reach at the point $Q(0, 2)$. The man is walking at a constant speed. If the man reaches the point Q in the minimum time, then $50((PR)^2 + (RQ)^2)$ is equal to _____. **01 Sep 2021 (E)**

Q66. If p and q are the lengths of the perpendiculars from the origin on the lines, $x \operatorname{cosec} \alpha - y \sec \alpha = k \cot 2\alpha$ and $x \sin \alpha + y \cos \alpha = k \sin 2\alpha$ respectively, then k^2 is equal to : **31 Aug 2021 (M)**

- (1) $2p^2 + q^2$ (2) $p^2 + 2q^2$
(3) $4p^2 + p^2$ (4) $4p^2 + q^2$

Q67. Let A be the set of all points (α, β) such that the area of triangle formed by the points $(5, 6)$, $(3, 2)$ and (α, β) is 12 square units. Then the least possible length of a line segment joining the origin to a point in A , is : **31 Aug 2021 (E)**

- (1) $\frac{8}{\sqrt{5}}$ (2) $\frac{12}{\sqrt{5}}$
(3) $\frac{16}{\sqrt{5}}$ (4) $\frac{4}{\sqrt{5}}$

Q68. Let $A(a, 0)$, $B(b, 2b + 1)$ and $C(0, b)$, $b \neq 0$, $|b| \neq 1$, be points such that the area of triangle ABC is 1 sq. unit, then the sum of all possible values of a is: **27 Aug 2021 (E)**

- (1) $\frac{-2b}{b+1}$ (2) $\frac{2b^2}{b+1}$
(3) $\frac{-2b^2}{b+1}$ (4) $\frac{2b}{b+1}$

Q69. Let A be a fixed point $(0, 6)$ and B be a moving point $(2t, 0)$. Let M be the mid-point of AB and the perpendicular bisector of AB meets the y -axis at C . The locus of the mid-point P of MC is _____ **27 Aug 2021 (M)**

- (1) $3x^2 + 2y - 6 = 0$ (2) $2x^2 - 3y + 9 = 0$
(3) $3x^2 - 2y - 6 = 0$ (4) $2x^2 + 3y - 9 = 0$

Q70. Let ABC be a triangle with $A(-3, 1)$ and $\angle ACB = \theta$, $0 < \theta < \frac{\pi}{2}$. If the equation of the median through B is $2x + y - 3 = 0$ and the equation of angle bisector of C is $7x - 4y - 1 = 0$, then $\tan \theta$ is equal to: **26 Aug 2021 (M)**

- (1) $\frac{3}{4}$ (2) $\frac{4}{3}$
(3) 2 (4) $\frac{1}{2}$

Q71. The point $P(a, b)$ undergoes the following three transformations successively:

- (a) reflection about the line $y = x$.

(b) translation through 2 units along the positive direction of x -axis.

(c) rotation through angle $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

If the co-ordinates of the final position of the point P are $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$, then the value of $2a + b$ is equal to:

27 Jul 2021 (E)

(1) 13

(2) 9

(3) 5

(4) 7

Q72. Two sides of a parallelogram are along the lines $4x + 5y = 0$ and $7x + 2y = 0$. If the equation of one of the diagonals of the parallelogram is $11x + 7y = 9$, then other diagonal passes through the point: **27 Jul 2021 (E)**

(1) (1, 2)

(2) (2, 2)

(3) (2, 1)

(4) (1, 3)

Q73*. Let the equation of the pair of lines, $y = px$ and $y = qx$, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is: **25 Jul 2021 (E)**

(1) $x^2 - 3xy + y^2 = 0$

(2) $x^2 + 4xy - y^2 = 0$

(3) $x^2 + 3xy - y^2 = 0$

(4) $x^2 - 3xy - y^2 = 0$

Q74. Consider a triangle having vertices $A(-2, 3)$, $B(1, 9)$ and $C(3, 8)$. If a line L passing through the circum-centre of triangle ABC , bisects line BC , and intersects y -axis at point $(0, \frac{\alpha}{2})$, then the value of real number α is _____. **20 Jul 2021 (E)**

Q75. Let the centroid of an equilateral triangle ABC be at the origin. Let one of the sides of the equilateral triangle be along the straight line $x + y = 3$. If R and r be the radius of circumcircle and incircle respectively of ΔABC , then $(R + r)$ is equal to : **18 Mar 2021 (E)**

(1) $\frac{9}{\sqrt{2}}$

(2) $7\sqrt{2}$

(3) $2\sqrt{2}$

(4) $3\sqrt{2}$

Q76. The equation of one of the straight lines which passes through the point $(1, 3)$ and makes an angles $\tan^{-1}(\sqrt{2})$ with the straight line, $y + 1 = 3\sqrt{2}x$ is **18 Mar 2021 (M)**

(1) $4\sqrt{2}x + 5y - (15 + 4\sqrt{2}) = 0$

(2) $5\sqrt{2}x + 4y - (15 + 4\sqrt{2}) = 0$

(3) $4\sqrt{2}x + 5y - 4\sqrt{2} = 0$

(4) $4\sqrt{2}x - 5y - (5 + 4\sqrt{2}) = 0$

Q77. The number of integral values of m so that the abscissa of point of intersection of lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer, is: **18 Mar 2021 (M)**

(1) 1

(2) 2

(3) 3

(4) 0

Q78. In a triangle PQR , the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x + y + 2 = 0$, then the centre of the circumcircle of the ΔPQR is: **17 Mar 2021 (M)**

(1) $(-1, 0)$

(2) $(-2, -2)$

(3) $(0, 2)$

(4) $(1, 4)$

Q79. Let $\tan \alpha$, $\tan \beta$ and $\tan \gamma$; $\alpha, \beta, \gamma \neq \frac{(2n-1)\pi}{2}$, $n \in N$ be the slopes of the three line segments OA , OB and OC , respectively, where O is origin. If circumcentre of $\triangle ABC$ coincides with origin and its orthocentre lies on y -axis, then the value of $\left(\frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma} \right)^2$ is equal to : 17 Mar 2021 (E)

Q80. Let $A(-1, 1)$, $B(3, 4)$ and $C(2, 0)$ be given three points. A line $y = mx$, $m > 0$, intersects lines AC and BC at point P and Q respectively. Let A_1 and A_2 be the areas of $\triangle ABC$ and $\triangle PQC$ respectively, such that $A_1 = 3A_2$, then the value of m is equal to : 16 Mar 2021 (E)

- (1) $\frac{4}{15}$ (2) 1
(3) 2 (4) 3

Q81. The intersection of three lines $x - y = 0$, $x + 2y = 3$ and $2x + y = 6$ is a/an 26 Feb 2021 (M)

- (1) Isosceles triangle (2) Equilateral triangle
(3) Right angled triangle (4) None of the above

Q82. The image of the point $(3, 5)$ in the line $x - y + 1 = 0$, lies on : 25 Feb 2021 (M)

- (1) $(x - 2)^2 + (y - 4)^2 = 4$ (2) $(x - 4)^2 + (y - 4)^2 = 8$
(3) $(x - 4)^2 + (y + 2)^2 = 16$ (4) $(x - 2)^2 + (y - 2)^2 = 12$

Q83*. If the curve $x^2 + 2y^2 = 2$ intersects the line $x + y = 1$ at two points P and Q , then the angle subtended by the line segment PQ at the origin is 25 Feb 2021 (E)

- (1) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{3}\right)$ (2) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{3}\right)$
(3) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$ (4) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$

Q84. A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axes is $\frac{1}{4}$. Three stones A , B and C are placed at the points $(1, 1)$, $(2, 2)$ and $(4, 4)$ respectively.

Then which of these stones is/are on the path of the man? 24 Feb 2021 (M)

- (1) C only (2) All the three
(3) B only (4) A only

Q85. Let a point P be such that its distance from the point $(5, 0)$ is thrice the distance of P from the point $(-5, 0)$. If the locus of the point P is a circle of radius r , then $4r^2$ (in the nearest integer) is equal to 24 Feb 2021 (E)

Q86. Let L denote the line in the xy -plane with x and y intercepts as 3 and 1 respectively. Then the image of the point $(-1, -4)$ in the line is : 06 Sep 2020 (E)

- (1) $\left(\frac{11}{5}, \frac{28}{5}\right)$ (2) $\left(\frac{29}{5}, \frac{8}{5}\right)$
(3) $\left(\frac{8}{5}, \frac{29}{5}\right)$ (4) $\left(\frac{29}{5}, \frac{11}{5}\right)$

Q87. A ray of light coming from the point $(2, 2\sqrt{3})$ is incident at an angle 30° on the line $x = 1$ at the point A . The ray gets reflected on the line $x = 1$ and meets x -axis at the point B . Then, the line AB passes through the point 06 Sep 2020 (M)

- (1) $\left(3, -\frac{1}{\sqrt{3}}\right)$ (2) $\left(4, -\frac{\sqrt{3}}{2}\right)$
(3) $\left(3, -\sqrt{3}\right)$ (4) $\left(4, -\sqrt{3}\right)$

Q88. If the line, $2x - y + 3 = 0$ is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines $4x - 2y + \alpha = 0$ and $6x - 3y + \beta = 0$ respectively, then the sum of all possible values of α and β is _____.

05 Sep 2020 (M)

Q89. If the perpendicular bisector of the line segment joining the points $P(1, 4)$ and $Q(k, 3)$ has y -intercept equal to -4 , then a value of k is;

04 Sep 2020 (E)

- (1) -2 (2) -4
(3) $\sqrt{14}$ (4) $\sqrt{15}$

Q90. A triangle ABC lying in the first quadrant has two vertices as $A(1, 2)$ and $B(3, 1)$. If $\angle BAC = 90^\circ$, and $\text{ar}(\Delta ABC) = 5\sqrt{5}$ sq. units, then the abscissa of the vertex C is :

04 Sep 2020 (M)

- (1) $1 + \sqrt{5}$ (2) $1 + 2\sqrt{5}$
(3) $2 + \sqrt{5}$ (4) $2\sqrt{5} - 1$

Q91. If a ΔABC has vertices $A(-1, 7)$, $B(-7, 1)$ and $C(5, -5)$, then its orthocentre has coordinates:

03 Sep 2020 (E)

- (1) $(-3, 3)$ (2) $(3, -3)$
(3) $(-\frac{3}{5}, \frac{3}{5})$ (4) $(\frac{3}{5}, -\frac{3}{5})$

Q92. The set of all possible values of θ in the interval $(0, \pi)$ for which the points $(1, 2)$ and $(\sin \theta, \cos \theta)$ lie on the same side of the line $x + y = 1$ is?

02 Sep 2020 (E)

- (1) $(0, \frac{\pi}{2})$ (2) $(\frac{\pi}{4}, \frac{3\pi}{4})$
(3) $(0, \frac{3\pi}{4})$ (4) $(0, \frac{\pi}{4})$

Q93. Let D be the centroid of the triangle with vertices $(3, -1)$, $(1, 3)$ and $(2, 4)$. Let P be the point of intersection of the lines $x + 3y - 1 = 10$ and $3x - y + 1 = 0$. Then, the line passing through the points D and P also passes through the point:

09 Jan 2020 (M)

- (1) $(-9, -6)$ (2) $(9, 7)$
(3) $(7, 6)$ (4) $(-9, -7)$

Q94. Let two points be $A(1, -1)$ and $B(0, 2)$. If a point $P(x', y')$ be such that the area of $\Delta PAB = 5$ sq. units and it lies on the line $3x + y - 4\lambda = 0$, then a value of λ is

08 Jan 2020 (M)

- (1) 4 (2) 3
(3) 1 (4) -3

Q95. Let $A(1, 0)$, $B(6, 2)$ and $C(\frac{3}{2}, 6)$ be the vertices of a triangle ABC . If P is a point inside the triangle ABC such that the triangles APC , APB and BPC have equal areas, then the length of the line segment PQ , where Q is the point $(-\frac{7}{6}, -\frac{1}{3})$, is

07 Jan 2020 (M)

Q96. The locus of the mid-points of the perpendiculars drawn from points on the line $x = 2y$, to the line $x = y$, is.

07 Jan 2020 (E)

- (1) $2x - 3y = 0$ (2) $5x - 7y = 0$
(3) $3x - 2y = 0$ (4) $7x - 5y = 0$

Q97. A triangle has a vertex at $(1, 2)$ and the mid points of the two sides through it are $(-1, 1)$ and $(2, 3)$. Then the centroid of this triangle is:

12 Apr 2019 (E)

- (1) $(\frac{1}{3}, 1)$
(3) $(\frac{1}{3}, 2)$

- (2) $(1, \frac{7}{3})$
(4) $(\frac{1}{3}, \frac{5}{3})$

Q98. A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is:

Note: In actual JEE Main paper, two options were correct for this question. Hence, we have changed one option.

12 Apr 2019 (E)

(1) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$

(2) $x + \sqrt{3}y = 8$

(3) $\sqrt{3}x + y = 8$

(4) $(\sqrt{3} - 1)x + \sqrt{3}y = 8\sqrt{2}$

Q99. Lines are drawn parallel to the line $4x - 3y + 2 = 0$, at a distance $\frac{3}{5}$ units from the origin. Then which one of the following points lies on any of these lines?

10 Apr 2019 (E)

- (1) $(\frac{1}{4}, -\frac{1}{3})$
(3) $(-\frac{1}{4}, -\frac{2}{3})$

- (2) $(-\frac{1}{4}, \frac{2}{3})$
(4) $(\frac{1}{4}, \frac{1}{3})$

Q100. Slope of a line passing through $P(2, 3)$ and intersecting the line $x + y = 7$ at a distance of 4 units from P , is

09 Apr 2019 (M)

- (1) $\frac{\sqrt{7}-1}{\sqrt{7}+1}$
(3) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$

- (2) $\frac{1-\sqrt{7}}{1+\sqrt{7}}$
(4) $\frac{1-\sqrt{5}}{1+\sqrt{5}}$

Q101. If the two lines $x + (a - 1)y = 1$ and $2x + a^2y = 1$, ($a \in R - \{0, 1\}$) are perpendicular, then the distance of their point of intersection from the origin is

09 Apr 2019 (E)

- (1) $\frac{2}{\sqrt{5}}$
(3) $\frac{2}{5}$

- (2) $\frac{\sqrt{2}}{5}$
(4) $\sqrt{\frac{2}{5}}$

Q102. A point on the straight line, $3x + 5y = 15$ which is equidistant from the coordinate axes will lie only in:

08 Apr 2019 (M)

- (1) 1st and 2nd quadrants
(3) 1st quadrant

- (2) 1st, 2nd and 4th quadrants
(4) 4th quadrant

Q103. Suppose that the points (h, k) , $(1, 2)$ and $(-3, 4)$ lie on the line L_1 . If a line L_2 passing through the points (h, k) and $(4, 3)$ is perpendicular to L_1 , then $\frac{k}{h}$ equals:

08 Apr 2019 (E)

- (1) $-\frac{1}{7}$
(3) 0

- (2) 3
(4) $\frac{1}{3}$

Q104. If the straight line $2x - 3y + 17 = 0$ is perpendicular to the line passing through the points $(7, 17)$ and $(15, \beta)$, then β equals :

12 Jan 2019 (M)

- (1) -5
(3) 5

- (2) $\frac{35}{3}$
(4) $-\frac{35}{3}$

Q105. If a straight line passing through the point $P(-3, 4)$ is such that its intercepted portion between the coordinate axes is bisected at P , then its equation is :

12 Jan 2019 (E)

(1) $4x + 3y = 0$

(3) $3x - 4y + 25 = 0$

(2) $4x - 3y + 24 = 0$

(4) $x - y + 7 = 0$

Q106. If in a parallelogram ABDC, the coordinates of A, B and C are respectively (1,2), (3,4) and (2, 5), then the equation of the diagonal AD is : 11 Jan 2019 (E)

(1) $5x - 3y + 1 = 0$

(3) $3x - 5y + 7 = 0$

(2) $5x + 3y - 11 = 0$

(4) $3x + 5y - 13 = 0$

Q107. In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y . If $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is 11 Jan 2019 (M)

(1) $\frac{3}{2}y$

(3) $\frac{c}{3}$

(2) $\frac{c}{\sqrt{3}}$

(4) $\frac{y}{\sqrt{3}}$

Q108. If the line $3x + 4y - 24 = 0$ intersects the x -axis at the point A and the y -axis at the point B , then the incentre of the triangle OAB , where O is the origin, is: 10 Jan 2019 (M)

(1) (4, 4)

(3) (4, 3)

(2) (3, 4)

(4) (2, 2)

Q109. Two vertices of a triangle are (0, 2) and (4, 3). If its orthocenter is at the origin, then its third vertex lies in which quadrant? 10 Jan 2019 (E)

(1) Fourth

(3) Third

(2) Second

(4) First

Q110. Two sides of a parallelogram are along the lines, $x + y = 3$ and $x - y + 3 = 0$. If its diagonals intersect at (2, 4), then one of its vertex is: 10 Jan 2019 (E)

(1) (3, 6)

(3) (2, 1)

(2) (2, 6)

(4) (3, 5)

Q111. A point P moves on the line $2x - 3y + 4 = 0$. If $Q(1, 4)$ and $R(3, -2)$ are fixed points, then the locus of the centroid of $\triangle PQR$ is a line: 10 Jan 2019 (M)

(1) with slope $\frac{2}{3}$ (3) parallel to y -axis(2) with slope $\frac{3}{2}$ (4) parallel to x -axis

Q112. Let the equations of two sides of a triangle be $3x - 2y + 6 = 0$ and $4x + 5y - 20 = 0$. If the orthocenter of this triangle is at (1, 1) then the equation of its third side is: 09 Jan 2019 (E)

(1) $122y + 26x + 1675 = 0$

(3) $26x + 61y + 1675 = 0$

(2) $26x - 122y - 1675 = 0$

(4) $122y - 26x - 1675 = 0$

Q113. Consider the set of all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true? 09 Jan 2019 (M)

(1) The lines are not concurrent.

(3) The lines are all parallel.

(2) The lines are concurrent at the point $(\frac{3}{4}, \frac{1}{2})$.

(4) Each line passes through the origin.

Q114. Let S be the set of all triangles in the xy -plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50 sq. units, then the number of elements in the set S is:

09 Jan 2019 (E)

(1) 36

(2) 32

(3) 9

(4) 18

ANSWER KEYS

1. (1)	2. (1)	3. (16)	4. (1)	5. (1)	6. (4)	7. (1)	8. (4)
9. (3)	10. (2)	11. (1)	12. (2)	13. (2)	14. (4)	15. (3)	16. (14)
17. (36)	18. (4)	19. (3)	20. (32)	21. (1)	22. (1)	23. (4)	24. (2)
25. (4)	26. (2)	27. (1)	28. (32)	29. (2)	30. (4)	31. (1)	32. (48)
33. (3)	34. (4)	35. (348)	36. (529)	37. (2)	38. (3)	39. (4)	40. (4)
41. (1)	42. (3)	43. (2)	44. (31)	45. (122)	46. (2)	47. (2)	48. (2)
49. (2)	50. (3)	51. (4)	52. (2)	53. (3)	54. (3)	55. (4)	56. (3)
57. (3)	58. (2)	59. (31)	60. (1)	61. (4)	62. (3)	63. (8)	64. (6)
65. (1250)	66. (4)	67. (1)	68. (3)	69. (4)	70. (2)	71. (2)	72. (2)
73. (3)	74. (9)	75. (1)	76. (1)	77. (2)	78. (2)	79. (144)	80. (2)
81. (1)	82. (1)	83. (3)	84. (3)	85. (56)	86. (1)	87. (3)	88. (30)
89. (2)	90. (2)	91. (1)	92. (1)	93. (1)	94. (2)	95. (5)	96. (2)
97. (3)	98. (1)	99. (2)	100. (2)	101. (4)	102. (1)	103. (4)	104. (3)
105. (2)	106. (1)	107. (2)	108. (4)	109. (2)	110. (1)	111. (1)	112. (2)
113. (2)	114. (1)						

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